Documentation

Write the solution documentation in docs/: how to run it, assumptions, decisions and complexity.

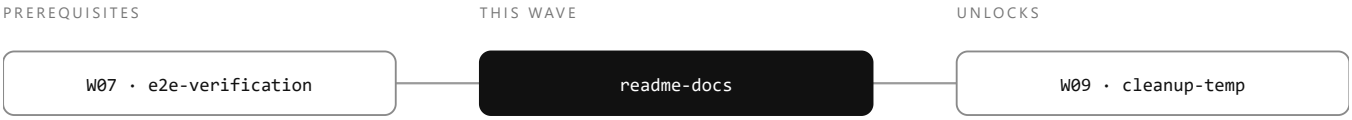
1 task | Requires Wave 07 | Unlocks Wave 09

Why this wave exists

Documentation comes after the end-to-end run so that it describes verified behaviour and not the plan.

How it fits

The diagram shows this wave's tasks, the earlier-wave tasks they depend on, and the later-wave tasks that depend on them.



Hand-off. Wave 9 removes temporary artefacts once the documentation is complete.

Tasks

Writing solution docs

`readme-docs`

Depends on: e2e-verification

Write two files in docs/. docs/solution.md covers how to run and test the solution and the verified results from wave 7. docs/decisions.md covers assumptions, retry and idempotency decisions, the per-key lease, partial failure handling, concurrency control, the production broker (Kafka, partitioned by stock key) and the time and space complexity of the matching. Do not overwrite the assignment's README.md.

`docs/solution.md · docs/decisions.md`

- [ ] Every claim matches the implemented code and the wave 7 evidence.
- [ ] Complexity is stated as: index build O(N) per country with N = 50,000; row match O(1) on average; bulk processing O(M) with M = 10,000.

Risks and decisions

- The assignment README.md is the mock environment guide. It is not overwritten.
- Every claim must match the implemented code and the wave 7 evidence.

Verification

```
npm run build
npm test
```

Done when

1. Every acceptance box is checked and the verification commands pass.
2. The session todos for this wave are marked done.
3. The final message includes a hand-off note for the next wave.

Agent brief for this wave: AGENT.md in the same folder.